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United States Patent	12404057
Kind Code	B2
Date of Patent	September 02, 2025
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Method of configuring air accelerator

Abstract

The method includes first configuring a first cross-sectional flow area of a throat within a first air accelerator to discharge a target quantity of a particulate material, the first air accelerator including a body and an adjustable member, an interior surface of the body defining a cavity containing the adjustable member, the throat being at an air outlet of a chamber defined between the body and the adjustable member, the throat being near a discharge end of a singular bore defined by the adjustable member, the first configuring including axially moving the adjustable member along a first longitudinal length of the cavity.

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Appl. No.:	18/194800
Filed:	April 03, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230249853 A1	Aug. 10, 2023

Related U.S. Application Data

continuation parent-doc US 13546649 20120711 US 9963253 20180508 child-doc US 15973860
division parent-doc US 17340184 20210607 US 11618596 child-doc US 18194800
division parent-doc US 15973860 20180508 US 11027860 20210608 child-doc US 17340184
us-provisional-application US 61506465 20110711

Publication Classification

Int. Cl.: **B65B37/14** (20060101); **B65B1/04** (20060101); **B65B1/30** (20060101); B65B9/02 (20060101)

U.S. Cl.:

CPC **B65B37/14** (20130101); **B65B1/04** (20130101); **B65B1/30** (20130101); B65B9/02 (20130101)

Field of Classification Search

CPC: B65B (37/14); B65B (1/04); B65B (1/30); B65B (9/02)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a divisional of U.S. patent application Ser. No. 17/340,184, filed Jun. 7, 2021, which is a divisional of U.S. patent application Ser. No. 15/973,860, filed May 8, 2018, which is a continuation application of U.S. patent application Ser. No. 13/546,649, filed Jul. 11, 2012, which claims priority under 35 U.S.C. § 119 (e) to U.S. Provisional Application No. 61/506,465, filed on Jul. 11, 2011, the entire contents of each which are incorporated herein by reference thereto.

FIELD OF THE DISCLOSURE

(1) This disclosure generally pertains to apparatus for metering material that includes particles. More specifically, this disclosure concerns apparatus having a compressed air acceleration.

OVERVIEW

(2) This disclosure has particular application to pouching machines used for forming and assembling pouches of particulate material, such as by way of example fine cut smokeless tobacco. Typical pouching machines simultaneously form and assemble, for example, ten pouches from a substantially continuous strip or web of pouch material and metered charges of prepared smokeless tobacco. To effect the simultaneous pouch assembly, pouching machines typically include a bank of generally vertical tobacco feed tubes. Typical pouching machines also include arrangements for drawing and directing a strip or ribbon of pouch web to each feed tube, and wrapping the strip around the corresponding feed tube to form a tubular formation, as well as arrangements to repetitively close and seal that tubular formation so as to form a lower transverse seam at a lower end portion of the tubular web formation just prior to charging each tubular formation with predetermined amount of smokeless tobacco. The pouching machine further includes arrangements for repetitively feeding individual charges of tobacco down corresponding feed tubes and into corresponding tubular formations. After each tobacco charge, the pouching machines close and seal the tubular formation at a second location above the tobacco charge to form an individual loaded

and sealed pouch, which is then severed from the tubular formation.

(3) Typically, smokeless tobacco material has a low moisture content, for example, about 30 to about 40% moisture level, and optionally includes flavorants, humectants and/or other tacky substances. Accordingly, smokeless tobacco has a tendency to stick to machine surfaces. Such smokeless tobacco is difficult to feed through pouch forming machines that rely merely on gravity feed techniques. Some pouching machinery incorporates pressurized air in the tobacco feed tubes to augment gravitational delivery of the smokeless tobacco charges. Because drier tobaccos are lighter than wetter tobaccos, the drier tobaccos have a greater tendency to scatter if subjected to jets of pressurized air during feeding, and that scatter can adversely affect the top seal on the associated pouch.

(4) Prior pouching machines include a tobacco feed mechanism for repetitively discharging a predetermined amount of tobacco from a hopper or the like into a funnel at the upper end portion of a tobacco feed tube. Generally, if gravity is the only active force to move the tobacco down the feed tube, a charge of tobacco released into the tube forms into a column of tobacco traveling down the feed tube such that it is constrained along a significant path length that may be too long for proper filling operations. More particularly, not all of the entrained tobacco may have time to enter the confines of a partially closed pouch before the machine closes and seals the pouch along its upper transverse seam.

(5) One solution has been to establish a Venturi arrangement at the base of the funnel. In this arrangement, pressurized air is introduced into the feed tube from a manifold through four to six or so small channels. Those small channels are fixed in size and may vary from tube to tube depending on machine tolerances and the like. Any clogging of one or more of the small channels tends to affect tobacco delivery for that feed tube in such a way that the bank of feed tubes performs inconsistently from one feed tube to another.

(6) Another disadvantage of the foregoing arrangement is that the small channels may impart a horizontal or transverse velocity component to the air being introduced through the small channels, with the result that some tobacco flow back may be caused.

(7) It is desired to have the feed tubes of the bank of tobacco feed tubes operate consistently amongst one another so that filling operations across the entire bank are consistent with one another.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The many innovative features and aspects of the present disclosure will be apparent to those skilled in the art when this specification is read in conjunction with the attached drawings wherein like reference numerals are applied to like elements and wherein:

(2) FIG. 1 is a schematic view in partial cross section of tobacco dosing apparatus;

(3) FIG. 1A is a partial cross-sectional view of the feed apparatus of FIG. 1;

(4) FIG. 2 is an enlarged, partial cross-sectional view taken through the dose delivery apparatus of FIG. 1;

(5) FIG. 3 is a detail view of the venturi discharge for the air accelerator unit of the dose delivery apparatus; and

(6) FIG. 4 is a schematic illustration of a calibration set-up.

DETAILED DESCRIPTION

(7) In the production of pouched products, including for example and without limitation, smokeless tobacco products, continuous-motion packaging machinery is often used, and is commonly known as form/fill/seal equipment. Such machinery receives packaging material in substantially continuous strips, receives material to be pouched as a substantially continuous supply from a

supply chamber, meters substantially uniform quantities of the material, partially forms a pouch, fills the metered material into the pouch, and finally seals the pouch such that the pouch surrounds that material. While various companies make such equipment, one such company is known as Ropak.

(8) Typical form/fill/seal equipment produces pouched products in a plurality of parallel streams of packaging material and product. For example, 5, 10, or more parallel lanes may be provided. Operating speeds on the order of 100 cycles per minute are known for each of the parallel lanes. As may be expected, that actual manufacturing speed depends on, for example, product flow characteristics, packaging materials used, and temperature at which filling occurs.

(9) In accord with this disclosure, a form/fill/seal apparatus **10** typically includes a plurality of suitable dose delivery apparatuses **20** (see FIG. **1**) to deliver particulate material in predetermined quantities. Typically, the form/fill/seal apparatus **10** receives a quantity of material to be parsed into predetermined quantities of doses of that material, and then delivers each predetermined quantity of material to a dose delivery apparatus **20**. The dose delivery apparatus **20** moves the predetermined quantity of material to a portion of the form/fill/seal apparatus where a pair of continuous webs **22**, **24** have been joined with a transverse seal **28** and longitudinal edge seals **26**, **26'** so as to define a pocket or pouch **29**. That pocket or pouch **29** is typically formed around the end **30** of a discharge tube of the dose delivery tube of a corresponding dose delivery apparatus **20**. Alternatively, a single web may be folded into a tubular form about the dose delivery tube and sealed along a single longitudinal edge, whereupon transverse seals applied to the tubular structure define a pouch **29**. Since the dose delivery apparatuses **20** are essentially identical, it will suffice to describe one in detail, with it being understood that the others are substantially the same. The principal difference from one dose apparatus **20** to another resides in its connection with the supply conduit.

(10) Each dosing apparatus **20** may include a supply conduit **24** connected at one end to the form/fill/seal apparatus **10** and connected at the other end to metering apparatus **12**. The metering apparatus **12** is operable to receive particulate material from the apparatus **10**, parse the particulate material into predetermined portions, doses, or quantities, and then deliver those predetermined portions, doses, or quantities of particulate material to the upper end of the dose delivery apparatus **20** at predetermined time intervals. The predetermined time intervals are selected so that a dose is delivered to the dose delivery apparatus **20** as each partial pouch is ready to be filled.

(11) While the metering apparatus **12** may take a variety of physical forms and arrangements, a presently preferred arrangement is depicted in FIG. **1**. More specifically, the metering apparatus **12** preferably includes a pair of generally parallel feed screws **14a**, **14b** that are arranged so as to be generally perpendicular to the axis of the dose delivery apparatus **20**. A suitable conventional drive mechanism is connected to at least one of the feed screws **14a**, **14b** such that the two feed screws rotate in the same direction about their respective axes. The drive mechanism is controlled, in a conventional manner, such that the feed screws intermittently rotate, with the time interval of the intermittent rotation being operable to define the predetermined dose of particulate tobacco material delivered to the dose delivery apparatus **20**.

(12) The feed screws **14a**, **14b** are preferably designed such that the flight of one screw cleans the flight of the adjacent screw as the two screws rotate. This characteristic of the feed screws **14a**, **14b** helps assure consistent weight and volume for the predetermined doses being delivered to the dose delivery apparatus **20**. Furthermore, the feed screws **14a**, **14b** are preferably fabricated from polyether ether ketone (PEEK).

(13) The metering apparatus **12** also includes a housing **16** (see FIG. **1A**) within which the feed screws **14a**, **14b** are positioned and within which those feed screws are mounted for rotation. The discharge end of the housing **16** is positioned above the inlet to the dose delivery apparatus **20**, and may be offset from both the center and the edge as depicted so that particulate tobacco material of a given dose can drop directly in to dose delivery apparatus **20**. The housing **16** closely conforms to the peripheral edge of the flight of each feed screw **14a**, **14b** so that particulate material does not

spill over the edge of the flight and dosing quantity is thus controlled. Preferably, the housing **16** is also fabricated from PEEK.

(14) The discharge end of the housing **16** is connected to a snout **18** which encloses the end of the housing and couples the housing **16** to the upper end of the funnel **32** of the dose delivery apparatus **20**. The snout **18** assures that particulate tobacco material delivered to the dose delivery apparatus **20** by the feed screws **14a**, **14b** does not escape and falls into the dose delivery apparatus **20**. In addition, the snout **18** is effective to avoid any external contamination of the particulate tobacco material passing therethrough. The snout **18** is also preferably fabricated from PEEK.

(15) The use of PEEK as a preferred material for fabrication of the feed screws **14a**, **14b**, the housing **16**, and the snout **18** has several advantageous and desirable attributes. PEEK functions as a thermal insulator. Thus, use of PEEK between the delivery apparatus **10** and the dose delivery apparatus **20** functions to substantially thermally insulated those apparatuses from one another. Moreover, PEEK substantially reduces and effectively avoids sticking of the particulate tobacco material to the surfaces of the housing, the feed screws, and the snout. Especially where the apparatus must be disassembled and cleaned on a regular basis (e.g., daily), this attribute is highly advantageous because it can reduce the cleaning time and thus add more processing time to the apparatus.

(16) For purposes of this disclosure, the particulate material may be particulate tobacco that has optionally been blended with other components including, for example, flavorants, humectants, and/or other substances, some or all of which may be tacky or may add tackiness to the particulate tobacco. The particulate tobacco material may include fine cut tobacco that has been comminuted at about 70 cuts per inch. Preferred particulate tobacco material may include up to about 39% oven volatiles.

(17) The snout **18** of the metering apparatus **12** attaches to a supply funnel **32** (see FIG. 1) at the inlet of the dose delivery assembly **20**. Preferably, the supply funnel **32** is circularly symmetric about an axis passing therethrough. At the bottom end of the supply funnel **32**, and in communication with the interior of the supply funnel, an air accelerator assembly **34** is provided. This air accelerator assembly **34** is operable to provide continuous or pulsed flow of particulate tobacco material. To that end, the air accelerator assembly **34** connects with an air supply conduit **38**, which in turn receives pressurized air from an air supply **40**. The air supply **40** may be a pump, air compressor, plenum chamber, or the like, as may be desired or appropriate for a particular application. A valve **42** may be in fluid communication with the air supply **40** and the air accelerator assembly **34**. As desired, the valve **42** may be operable to interrupt air flow to the air accelerator assembly **34** so as to start, stop, and/or pulse air delivered to the air accelerator assembly **34**. Typically, air at ambient temperature and pressure in the range of 4-18 psig has been found to be suitable for use with an air accelerator assembly **34** of the type described herein.

(18) At the bottom end, the air accelerator assembly **34** attaches to a dosing tube **36**. That dosing tube **36** preferably terminates in a position where the pouch has been partially formed and can receive particulate material from the discharge end of the dosing tube **36**.

(19) The air accelerator assembly **34** includes a body **50**, and an internal member **52** which is axially adjustable with respect to the body **50** along an axis **54**. Preferably, the funnel member **32** is rotationally symmetric about the axis **54**. Internal surfaces of the body **50** that are exposed to air flow, as well as surfaces of the internal member **52** that are exposed to air flow or to product flow are also rotationally symmetric with respect to the axis **54**.

(20) The narrow or lower end of the funnel member **32** preferably includes a radially extending flange **56** having a periphery that corresponds to the outer peripheral surface of the body **50**. In addition, the flange **56** of the funnel member **32** includes a radially extending annular face **64** which is configured to mate with a corresponding radially extending annular face **66** at the upper end of the body **50**. The flange **56** preferably also includes a projecting land **68** which is received in a threaded bore **70** of the body **50**. Cooperation between the projecting land **68** and the associated

bore **70** assures that the body **50** and the funnel member **32** are coaxial when joined together. To that end, a plurality of axially extending bolts, or threaded fasteners **58**, may be used to attach the flange **56** and the body **50**. Suitable gasket material may be provided between the abutting surfaces **64**, **66** of the flange **56** and the body **50**, respectively, if desired.

(21) Extending longitudinally through the body **50**, along the axis **54**, is a body cavity that includes a threaded, generally cylindrical portion adjacent the funnel member **32**, a frustoconical portion **72** extending downstream from the threaded portion, and a discharge tube connection portion at the lower or bottom end of the body **50**. The frustoconical portion **72** essentially matches the diameter of the threaded portion at its upstream end. In addition, the downstream or lower end of the frustoconical portion **72** is preferably sized to have a diameter corresponding to the inside diameter of the discharge tube **36**. The discharge tube **36** is preferably attached to the downstream end of the body **50** using a suitable conventional attachment. For example, any of a threaded connection, a welded connection, or an adhesively bonded and sealed connection would be satisfactory.

(22) Turning to the longitudinally movable or adjustable member **52** of the air accelerator assembly **34**, the adjustable member **52** includes a generally cylindrical longitudinal bore **80** extending from the upstream end to the downstream end of the adjustable member **52**. The longitudinal bore **80** preferably has a diameter corresponding to the opening at the discharge end of the funnel member **32** so that particulate material can move downwardly through the funnel member **32** and into the longitudinal bore **80** substantially free of impediment.

(23) The upper or upstream end of the adjustable member **52** includes a flange portion **84** preferably having a peripherally threaded portion that mates with the threaded portion of the cavity in the body **50**. Cooperation between the externally threaded flange **84** and the internally threaded portion of the body cavity not only secures the adjustable member **52** in the body **50**, but also allows the adjustable member **52** to have its spatial relationship with the body **50** controlled in the longitudinal direction along the axis **54**.

(24) Preferably, the exterior surface of the adjustable member **52** also includes a frustoconical surface **82** extending from the flange **84** to the distal end **88** at the downstream end of the adjustable member **52**. Preferably, the frustoconical surface **82** meets the longitudinal bore **80** at the distal end **88** of the adjustable member **52** so that an acute sharp angle is defined in the material of the adjustable member **52**. Both the frustoconical surface **82** of the adjustable member **52** and the frustoconical portion of the cavity in the body **50** are preferably polished. Because the facing frustoconical surfaces define a chamber for pressurized air, and because it is desirable to accurately control the flow rate of pressurized air through that chamber, it is believed to be important that those facing frustoconical surfaces be as smooth as possible so as to avoid creating inconsistent resistance to air flow from one air accelerator assembly **34** to another. Accordingly, these facing frustoconical surfaces may be honed and/or polished so that the surface roughness is about 100 microinches or less, and preferably about 30 microinches or less.

(25) As noted, the cavity of the body **50** and the frustoconical surface **82** of the adjustable member **52** cooperate to define a chamber **90** for pressurized air. That chamber **90** has fluid communication with the conduit **38**, and thus the pump **40** and associated control valve **42** (see FIG. 1). The frustoconical surface **82** (see FIG. 3) of the adjustable member defines an angle a with the axis **54** of its central bore **80**. The frustoconical surface portion **72** of the cavity in the body **50** has an angle b with the axis **54**. The distal end **88** of the adjustable member **52** cooperates with the frustoconical surface portion **72** of the cavity in the body **50** to define a throat or minimum flow area at the downstream end of the chamber **90**. To assure that the flow area through the chamber **90** decreases as air moves downstream toward the throat, the angle a must be less than the angle b . Thus, the chamber **90** (see FIG. 3) effectively comprises a venturi through which pressurized air in the chamber **90** passes as it moves toward and through the reduced area throat **100** (outlet of the chamber **90**). With the longitudinal adjustability of the member **52** in the direction of the arrow **102**, the throat **100** can be adjusted as described more fully below to calibrate and adjust the

various air acceleration assemblies of a form/fill/seal machine.

(26) Since it is also important that air supplied to the chamber **90** (see FIG. 2) through the conduit **38** be constrained to pass out of the chamber **90** only through the throat **100**, a suitable conventional gasket **86** may be provided at the upper end of the chamber **90** between the flange **84** of the adjustable member **52** and the cavity of the body **50**.

(27) In a preferred embodiment, the body **50** and the adjustable member **52** are constructed from air-hardened tool steel.

(28) As noted above, the particulate tobacco material processed through the dosing tube assembly described above may exhibit tackiness. Accordingly, one or more of the interior surface of the funnel member **32**, the cylindrical channel **80** of the adjustable member **52**, and the interior of the discharge tube **36** may also be coated with polyether ether ketone (PEEK). More preferably, the adjustable member **52** may be constructed entirely from PEEK. Such a coating can improve mechanical and chemical resistance to the particulate material as that material moves through the dosing tube assembly.

(29) It will now be understood by those skilled in the art that the tapered angle b of the adjustable member **52** (see FIG. 2) is greater than the corresponding taper angle a of the frustoconical channel of the body **50** such that as the member **52** is threaded into the body **50** a tapered convergent chamber **80** is defined around a portion of the adjustable member **52** in the space provided between the body **50** and the member **52**. As the member **52** is threaded further and further into the body **50**, the annular discharge orifice or throat **100** at the distal end **88** of the member **52**, and near the base of the body **50**, becomes smaller and smaller.

(30) Conventional set screws may be provided as a locking means to fix or otherwise lock the relative positions of the member **52** and the body **50**.

(31) To prepare an air acceleration assembly **34** for use, the assembly **34** and its discharge tube **36** are removed from the tobacco feed system. Then the assembly **34** is calibrated by adjusting the throat of the variable venturi such that a predetermined force is obtained from the associated discharge tube. To that end, the assembly **34** with its discharge tube **36** is positioned in a fixture such that the end **36** at the base of the discharge tube **36** is proximately positioned relative to a suitable conventional a precision scale **112**. The discharge tube **36** is held at a predetermined stand-off distance d above the surface of the precision scale **112**. Preferably that predetermined stand-off distance d between the end of the discharge tube **36** and the precision scale **112** is about 1 mm.

(32) The feed tube is connected to the source **40** of pressurized air through the conduit **38** (see FIG. 1) and the pressure regulator **42**. The pressure regulator is adjusted to a desired operating pressure for the tobacco pouching machine, for example eighteen psig. Then the longitudinally adjustable member **52** is rotated so that it can be adjusted either up or down relative to the body **50** until the discharge of air through the discharge tube onto the precision scale registers a reading of a predetermined force, preferably in the range of about 20 to about 30 g. For example, the predetermined force or target scale reading might be 25 g. Once body **50** and member **52** have been adjusted so that the desired force reading is obtained, the member **52** is locked in place relative to the body **50** by a set screw or other suitable mechanism to fix the relative position of the body **50** and the member **52**. While a mechanical locking arrangement such as a set screw may be used, the relative positions of the member **52** and the body **50** are most preferably permanently attached to one another, as by welding, so that the calibration is fixed. Otherwise, when the feed tube is cleaned (typically a daily occurrence), recalibration is required. The foregoing steps are repeated for each remaining air acceleration assembly **34** until all assemblies **34** have been calibrated to provide the same predetermined force.

(33) After each air acceleration assembly **34** has been calibrated and returned to the tobacco feed mechanism, the pouching machine, i.e., the form/fill/seal machine, is ready for operation. Typically, a machine operator adjusts the air regulator **42** (FIG. 1) of the pouching machine to achieve desired pouch loading operation across the bank of feed tubes.

(34) At one extreme, the air pressure may be too high, in which case the tobacco is driven into the pouch with such force that the pouch tends to open or cause tobacco to enter the first lower transverse seal of the pouch being formed. In another case, the pressure may be too low such that the upper transfer seam is closed and sealing initiated before all the tobacco has fully arrived into the body portion of the pouch. For this latter condition, the operator typically increases the operating pressure. Once the filling sequence has been optimized, the operator is assured uniform filling across the bank of feed tubes, because each air acceleration assembly has been calibrated the same way.

(35) Preferably, the operating pressure of all feed lanes (or delivery apparatuses **20**) is adjustable from a single, common regulator **42**. Such arrangement contributes uniform tobacco feeding characteristics across the entire bank of feed lanes to enhance machine operation and performance. The arrangement assures that downstream timing requirements are uniformly met. For example the cutting knives for severing fully formed pouches operate uniformly at a fixed rate across the entire bank of feed lanes. The feed system as taught herein, with its locking down each air delivery system to a common, uniform calibration and uniform adjustment of operating pressure from a common regulator assures that tobacco is delivered at the right time and at the right speed across the bank of feed lanes. During operations, should delivery speed of the feed lanes drift, the operator may return the entire bank of feed lanes back into desired delivery speed by observing a single feed lane while adjusting the common regulator.

(36) In this description, the word “substantially” is used as an adjective to show that the modified term need not be used literally, but is intended to include equivalent terms which do not materially depart from the spirit and scope of the term. When the word “substantially” is used in connection with a geometric term, it is intended that the geometric term not be interpreted rigidly with respect to geometric definitions.

(37) To similar effect, the word “about” is used in this description in connection with numerical terms to demonstrate that mathematical precision is not required and that a tolerance of $\pm 10\%$ around that numerical term is intended.

(38) It will now be apparent to those skilled in the art that this specification provides a novel and unobvious improvement to a metering device for particulate material, particularly where pressurized fluid functions to assist movement of the particulate material through the apparatus. Furthermore, it will be apparent to those skilled in the art that numerous modifications, variations, substitutions, and legal equivalents exist for features of the invention described herein. Accordingly, it is expressly intended that all such modifications, variations, substitution, and legal equivalents that fall within the spirit and scope of the appended claims be embraced thereby.

Claims

1. A method, comprising: first configuring a first cross-sectional flow area of a throat within a first air accelerator to discharge a target quantity of a particulate material, the first air accelerator including a body and an adjustable member, an interior surface of the body defining a cavity containing the adjustable member, the throat being at an air outlet of a chamber defined between the body and the adjustable member, the throat being near a discharge end of a singular bore defined by the adjustable member, the first configuring including axially moving the adjustable member along a first longitudinal length of the cavity, the singular bore having a uniform cross-sectional flow area along a longitudinal length of the singular bore.
2. The method of claim 1, wherein the first configuring further includes first establishing a test airflow through the chamber, measuring a first force of the test airflow exiting the first air accelerator, and determining a first axial position of the adjustable member that causes the first force to equal a target force, the target force being a desired force that causes the first air accelerator to deliver the target quantity of the particulate material, the chamber being defined in

part by an external frustoconical surface of the adjustable member and internal frustoconical surface that is part of the interior surface, the internal frustoconical surface having a first angle of incline that is steeper than a second angle of incline of the external frustoconical surface relative to a first longitudinal centerline running through a center of the singular bore.

3. The method of claim 2, further comprising: second establishing an operating airflow through the chamber with the adjustable member in the first axial position; third establishing a flow of the particulate material into the singular bore; and discharging the target quantity of the particulate material from the singular bore with assistance from the operating airflow.

4. The method of claim 1, further comprising: first establishing an operating airflow through the chamber; second establishing a flow of the particulate material into the singular bore; and discharging the target quantity of the particulate material from the singular bore with assistance from the operating airflow.

5. The method of claim 4, wherein the second establishing further includes connecting a funnel of a metering assembly to an entrance end of the body such that the funnel is in communication with the singular bore, a first internal diameter of an outlet of the funnel being equal to a second internal diameter of the singular bore, and configuring the metering assembly to deliver the flow of the particulate material through the funnel and into the singular bore.

6. The method of claim 4, wherein the second establishing further includes connecting a funnel of a metering assembly to an entrance end of the body such that the funnel is in communication with the singular bore, a first internal diameter of an outlet of the funnel being equal to a second internal diameter of the singular bore, and configuring parallel feed screws of the metering assembly to transport the flow of the particulate material in a horizontal direction into and through the funnel and into the singular bore, the funnel having a first longitudinal centerline running through a center of the funnel that is positioned vertically and is aligned with a force of gravity, the first longitudinal centerline being aligned with a second longitudinal centerline running through a center of the singular bore.

7. A method, comprising: first configuring a first cross-sectional flow area of a throat within a first air accelerator to discharge a target quantity of a particulate material, the first air accelerator including a body and an adjustable member, an interior surface of the body defining a cavity containing the adjustable member, the throat being at an air outlet of a chamber defined between the body and the adjustable member, the throat being near a discharge end of a singular bore defined by the adjustable member, the first configuring including axially moving the adjustable member along a first longitudinal length of the cavity; first establishing an operating airflow through the chamber; second establishing a flow of the particulate material into the singular bore; discharging the target quantity of the particulate material from the singular bore with assistance from the operating airflow; inserting a dosing tube into a first end of a partially formed pouch, the singular bore being in communication with the dosing tube; discharging the target quantity of the particulate material from the dosing tube into the partially formed pouch; and sealing the first end.

8. A method, comprising: first configuring a first cross-sectional flow area of a throat within a first air accelerator to discharge a target quantity of a particulate material, the first air accelerator including a body and an adjustable member, an interior surface of the body defining a cavity containing the adjustable member, the throat being at an air outlet of a chamber defined between the body and the adjustable member, the throat being near a discharge end of a singular bore defined by the adjustable member, the first configuring including axially moving the adjustable member along a first longitudinal length of the cavity; first establishing an operating airflow through the chamber; second establishing a flow of the particulate material into the singular bore; and discharging the target quantity of the particulate material from the singular bore with assistance from the operating airflow, wherein the second establishing further includes connecting a funnel of a metering assembly to an entrance end of the body such that the funnel is in communication with the singular bore, a first internal diameter of an outlet of the funnel being equal to a second internal

diameter of the singular bore, and configuring the metering assembly to deliver the flow of the particulate material through the funnel and into the singular bore.

9. The method of claim 8, further comprising: inserting a dosing tube into a first end of a partially formed pouch, the singular bore being in communication with the dosing tube, a third internal diameter of the dosing tube being equal to the second internal diameter; discharging the target quantity of the particulate material from the dosing tube into the partially formed pouch; and sealing the first end.

10. A method, comprising: first configuring a first cross-sectional flow area of a throat within a first air accelerator to discharge a target quantity of a particulate material, the first air accelerator including a body and an adjustable member, an interior surface of the body defining a cavity containing the adjustable member, the throat being at an air outlet of a chamber defined between the body and the adjustable member, the throat being near a discharge end of a singular bore defined by the adjustable member, the first configuring including axially moving the adjustable member along a first longitudinal length of the cavity, wherein the first configuring further includes first establishing a test airflow through the chamber, measuring a first force of the test airflow exiting the first air accelerator, and determining a first axial position of the adjustable member that causes the first force to equal a target force, the target force being associated with the target quantity of the particulate material, the chamber having a continuously smaller cross-sectional flow area from an air inlet to the air outlet of the chamber.

11. The method of claim 10, further comprising: second establishing an operating airflow through the chamber with the adjustable member in the first axial position; third establishing a flow of the particulate material into the singular bore; and discharging the target quantity of the particulate material from the singular bore with assistance from the operating airflow.

12. The method of claim 11, further comprising: repeating the first establishing, the measuring and the determining steps on a periodic basis, following the discharging of the target quantity for a period of time.

13. The method of claim 11, further comprising: repeating the first configuring step for a plurality of additional air accelerators, each of the plurality of additional air accelerators being identical to the first air accelerator, wherein the second establishing establishes the operating airflow by connecting a common air supply to each respective one of the chambers of each one of the plurality of additional air accelerators.

14. The method of claim 10, further comprising: inserting a gasket at an upper end of the chamber at an interface between the body and the adjustable member; and connecting a flange to an upper end of the first air accelerator, a projecting land of the flange extending into the cavity, an annular gap existing between the projecting land and the adjustable member.

15. The method of claim 14, further comprising: second establishing an operating airflow through the chamber with the adjustable member in the first axial position; third establishing a flow of the particulate material into the singular bore; and discharging the target quantity of the particulate material from the singular bore with assistance from the operating airflow.

16. The method of claim 14, further comprising: fixing the adjustable member in the first axial position prior to connecting the flange.

17. A method, comprising: first configuring a first cross-sectional flow area of a throat within a first air accelerator to discharge a target quantity of a particulate material, the first air accelerator including a body and an adjustable member, an interior surface of the body defining a cavity containing the adjustable member, the throat being at an air outlet of a chamber defined between the body and the adjustable member, the throat being near a discharge end of a singular bore defined by the adjustable member, the first configuring including axially moving the adjustable member along a first longitudinal length of the cavity, wherein the first configuring further includes first establishing a test airflow through the chamber, measuring a first force of the test airflow exiting the first air accelerator, and screwing the adjustable member into a first axial position using

first threads on an upper end of the adjustable member that mate with second threads on an upper end of the interior surface, the first axial position causing the first force to equal a target force, the target force being a desired force that causes the first air accelerator to deliver the target quantity of the particulate material, the chamber having a continuously smaller cross-sectional flow area from an air inlet to the air outlet of the chamber.

18. The method of claim 17, further comprising: second establishing an operating airflow through the chamber with the adjustable member in the first axial position; third establishing a flow of the particulate material into the singular bore; and discharging the target quantity of the particulate material from the singular bore with assistance from the operating airflow.

19. The method of claim 1, wherein the particulate material is smokeless tobacco.
